

Problems and Solutions - Part 2

nonlinear equations

numerical integration

finite-difference method for boundary-value problems

nonlinear equations

Concepts you need to know

- nonlinear problems
- iterative methods
- convergence order of an iterative method
- stopping criteria: error, saturation, residual
- Bisection, Picard's and Newton-Raphson methods
- Newton-Raphson method for systems of nonlinear equations, Jacobian matrix
- when each of the methods converges or fails to converge

Questions you may get

- given a nonlinear problem, write it in standard and fixed-point forms
- write down the residual r_n or $\mathbf{r}_n$ of the problem at the iteration n
- compute one iteration of a method (Bisection, Picard's, Newton-Raphson)

Example 1

Question. Given the nonlinear problem:

$$\frac{\sin^2(x)}{1+x^2} = e^{-x^2},$$

write down the standard and fixed-point formulations and the residuals at the n -th iteration of an iterative method, which produces the sequence $p_0, \dots, p_n, \dots$

Solution.

Answer: standard formulation $f(x) = 0$

$$\frac{\sin^2(x)}{1+x^2} - e^{-x^2} = 0$$

Answer: fixed-point formulation $g(x) = x$

$$\frac{\sin^2(x)}{1+x^2} - e^{-x^2} + x = x$$

Example 1

Answer: residual in standard formulation $r_n = f(p_n)$

$$r_n = \frac{\sin^2(p_n)}{1 + p_n^2} - e^{-p_n^2}$$

Answer: residual in fixed-point formulation $r_n = g(p_n) - p_n$

$$\begin{aligned} r_n &= \frac{\sin^2(p_n)}{1 + p_n^2} - e^{-p_n^2} + p_n - p_n \\ &= \frac{\sin^2(p_n)}{1 + p_n^2} - e^{-p_n^2} \end{aligned}$$

Example 2

Question. Compute one iteration of Picard's en Newton-Raphson methods for the problem:

$$\begin{cases} 2x_1 - x_2 + x_1^2 = \frac{1}{9}, \\ -x_1 + 2x_2 + x_2^2 = \frac{13}{9}, \end{cases}$$

starting with the initial guess $x_1 = 0, x_2 = 0$.

Solution. For Picards' method the problem must be written in a vector fixed-point form $\underline{g}(\underline{p}) = \underline{p}$, where $\underline{p} = [x_1, x_2]^T$. For example,

$$\begin{bmatrix} 2x_1 - x_2 + x_1^2 - \frac{1}{9} + x_1 \\ -x_1 + 2x_2 + x_2^2 - \frac{13}{9} + x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$
$$\begin{bmatrix} 3x_1 - x_2 + x_1^2 - \frac{1}{9} \\ -x_1 + 3x_2 + x_2^2 - \frac{13}{9} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Example 2

Picard's iteration:

$$\underline{p}_1 = \underline{g}(\underline{p}_0)$$

In this case:

$$\underline{p}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$\underline{p}_1 = \begin{bmatrix} 3(0) - (0) + (0)^2 - \frac{1}{9} \\ -(0) + 3(0) + (0)^2 - \frac{13}{9} \end{bmatrix} = \begin{bmatrix} -\frac{1}{9} \\ -\frac{13}{9} \end{bmatrix}$$

Answer: Picard's iteration

$$\underline{p}_1 = \begin{bmatrix} -\frac{1}{9} \\ -\frac{13}{9} \end{bmatrix}$$

Example 2

For the Newton-Raphson method the problem must be written in a standard vector form $\underline{f}(\underline{p}) = \underline{0}$:

$$\begin{bmatrix} 2x_1 - x_2 + x_1^2 - \frac{1}{9} \\ -x_1 + 2x_2 + x_2^2 - \frac{13}{9} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

One iteration of the Newton-Raphson method is defined as

$$\begin{aligned} \underline{p}_0 &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \underline{p}_1 &= \underline{p}_0 - [J(\underline{p}_0)]^{-1} \underline{f}(\underline{p}_0) \end{aligned}$$

Compute Jacobian at $\underline{x}$:

$$J(\underline{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2 + 2x_1 & -1 \\ -1 & 2 + 2x_2 \end{bmatrix}$$

Example 2

Compute Jacobian at $\underline{p}_0$:

$$J(\underline{p}_0) = \begin{bmatrix} 2 + 2(0) & -1 \\ -1 & 2 + 2(0) \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Invert $J(\underline{p}_0)$:

$$[J(\underline{p}_0)]^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}^{-1} = \frac{1}{2(2) - (-1)(-1)} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Compute $\underline{f}$ at $\underline{p}_0$:

$$\underline{f}(\underline{p}_0) = \begin{bmatrix} 2(0) - (0) + (0)^2 - \frac{1}{9} \\ -(0) + 2(0) + (0)^2 - \frac{13}{9} \end{bmatrix} = \begin{bmatrix} -\frac{1}{9} \\ -\frac{13}{9} \end{bmatrix}$$

Example 2

Compute the iteration:

$$\underline{p}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\underline{p}_1 = \underline{p}_0 - [J(\underline{p}_0)]^{-1} \underline{f}(\underline{p}_0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -\frac{1}{9} \\ -\frac{13}{9} \end{bmatrix} = \begin{bmatrix} \frac{5}{9} \\ 1 \end{bmatrix}$$

Answer: Newton-Raphson iteration

$$\underline{p}_1 = \begin{bmatrix} \frac{5}{9} \\ 1 \end{bmatrix}$$

numerical integration

Concepts you need to know

- idea of numerical integration, general quadrature formula
- partition, splitting the integral into pieces
- common integration rules: left-/right-rectangle, midpoint, trapezoidal, Simpson's
- errors of common integration rules
- composite rule, error of a composite rule
- interpolatory quadratures and their errors

Questions you may get

- given an integral and an integration rule, determine the step size of the partition for which the error of numerical integration is below a given value
- describe the main differences between the Newton-Cotes, Clenshaw-Curtis and Gauss-Legendre quadratures

Example 1

Question. Consider the integral

$$\mathcal{I} = \int_0^{\pi} \sin^2(x) dx$$

How small should be the step h of the uniform partition so that the composite midpoint rule gives the error below ϵ . The error of the composite midpoint rule is

$$\left| \sum_{k=1}^n I_k - \int_a^b f(x) dx \right| \leq \frac{1}{24} M_2 (b-a) h^2, \quad M_2 = \max_{x \in [a,b]} |f^{(2)}(x)|$$

Solution.

$$\begin{aligned} \left| \sum_{k=1}^n I_k - \int_a^b f(x) dx \right| &< \epsilon \\ \frac{1}{24} M_2 (b-a) h^2 &< \epsilon \end{aligned}$$

Example 1

We achieve this by

$$\frac{1}{24} \max_{x \in [0, \pi]} |[\sin^2(x)]''| (\pi - 0) h^2 < \epsilon$$

$$\frac{\pi h^2}{24} \max_{x \in [0, \pi]} |2(\cos^2(x) - \sin^2(x))| < \epsilon$$

$$\frac{\pi h^2}{24} \max_{x \in [0, \pi]} |2 \cos(2x)| < \epsilon$$

$$\frac{\pi h^2}{24} 2 < \epsilon$$

$$h^2 < \frac{12\epsilon}{\pi}$$

Answer

$$h < \sqrt{\frac{12\epsilon}{\pi}}$$

finite-difference method for boundary-value problems

Concepts you need to know

- difference between initial-value and boundary-value problems
- types of boundary conditions: Dirichlet, Neumann, Robin
- uniform mesh over interval $[a, b]$, inner and boundary mesh-points
- $\mathcal{O}(h^2)$ approximations of u'' and u' at inner points
- $\mathcal{O}(h)$ and $\mathcal{O}(h^2)$ approximations of u' at boundary points
- relation between the number of unknowns and number of mesh-points (depending on boundary conditions)

Open questions you will get

- Write down the finite-difference equations approximating a given boundary-value problem on a uniform mesh dividing the interval $[a, b]$ into N elements
- Assemble the finite-difference equations into a linear algebraic system $A\mathbf{u} = \mathbf{f}$, provide matrix A and vectors $\mathbf{u}$, $\mathbf{f}$

Example Question: BVP for convection-diffusion equation

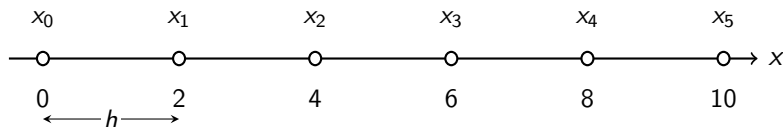
Consider the boundary-value problem:

$$\begin{aligned} -2u'' + 3u' &= x^2, \quad x \in (0, 10), \\ -2u'(0) &= 1, \quad u(10) = 1, \end{aligned}$$

with the unknown function $u(x)$. We introduce a uniform mesh $x_i = hi$, $i = 0, \dots, N$, obtained by dividing the interval $[0, 10]$ into $N = 5$ elements. Using $\mathcal{O}(h^2)$ finite-difference approximations at inner mesh-points and $\mathcal{O}(h)$ approximations at boundary points:

- 1 write down the finite-difference equations for the boundary points and for the inner point x_3
- 2 assemble the complete linear system $A\mathbf{u} = \mathbf{f}$ and write down A , $\mathbf{u}$, and $\mathbf{f}$

Draw and understand your uniform mesh



Mesh Properties

- Step size $h = (10 - 0)/5 = 2$
- Number of mesh-points: $5 + 1 = 6$
- Boundary points: $x_0 = 0$ and $x_5 = 10$
- Inner points: x_1, x_2, x_3, x_4

Understand your given values and unknowns

Use the common notation: $u(x_i) = u_i$

- Values of x : $x_0 = 0, x_1 = 2, x_3 = 4, \dots, x_4 = 8, x_5 = 10$
- Unknown values of $u(x)$: u_0, u_1, u_2, u_3, u_4
- Known values of $u(x)$: $u_5 = 1$
- Known values of the source function $f(x) = x^2$: $f_0 = x_0^2 = 0, f_1 = x_1^2 = 4, \dots, f_4 = x_4^2 = 64$

Recall the necessary FD approximations

- At inner mesh-points x_i :

$$u'_i = \frac{u_{i+1} - u_{i-1}}{2h} + \mathcal{O}(h^2)$$

$$u''_i = \frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} + \mathcal{O}(h^2)$$

- At boundary points x_0 and x_N :

$$u'_0 = \frac{u_1 - u_0}{h} + \mathcal{O}(h)$$

$$u'_N = \frac{u_N - u_{N-1}}{h} + \mathcal{O}(h)$$

Answer 1: write down the required FD equations

- At boundary points x_0 and x_5 :

$$-2 \frac{u_1 - u_0}{2} = 1$$
$$u_5 = 1$$

- At inner mesh-point x_3 :

$$-2 \frac{u_2 - 2u_3 + u_4}{4} + 3 \frac{u_4 - u_2}{4} = 36$$

Write down all FD equations

$$-2 \frac{u_1 - u_0}{2} = 1$$

$$-2 \frac{u_0 - 2u_1 + u_2}{4} + 3 \frac{u_2 - u_0}{4} = 4$$

$$-2 \frac{u_1 - 2u_2 + u_3}{4} + 3 \frac{u_3 - u_1}{4} = 16$$

$$-2 \frac{u_2 - 2u_3 + u_4}{4} + 3 \frac{u_4 - u_2}{4} = 36$$

$$-2 \frac{u_3 - 2u_4 + u_5}{4} + 3 \frac{u_5 - u_3}{4} = 64$$

$$u_5 = 1$$

Simplify and eliminate known values

$$u_0 - u_1 = 1$$

$$-\frac{5}{4}u_0 + u_1 + \frac{1}{4}u_2 = 4$$

$$-\frac{5}{4}u_1 + u_2 + \frac{1}{4}u_3 = 16$$

$$-\frac{5}{4}u_2 + u_3 + \frac{1}{4}u_4 = 36$$

$$-\frac{5}{4}u_3 + u_4 = 64 - \frac{1}{4}$$

Answer 2: write down the linear system $A\mathbf{u} = \mathbf{f}$

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ -\frac{5}{4} & 1 & \frac{1}{4} & 0 & 0 \\ 0 & -\frac{5}{4} & 1 & \frac{1}{4} & 0 \\ 0 & 0 & -\frac{5}{4} & 1 & \frac{1}{4} \\ 0 & 0 & 0 & -\frac{5}{4} & 1 \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 16 \\ 36 \\ 64 - \frac{1}{4} \end{bmatrix}$$